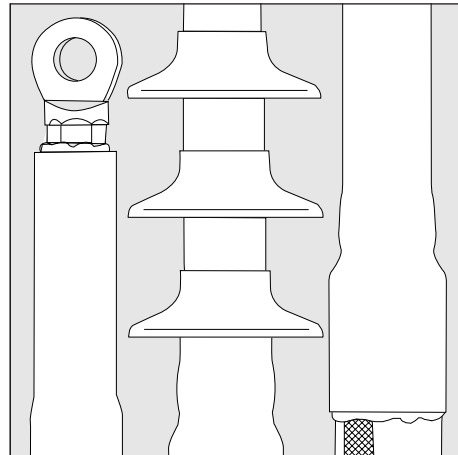




TE's Raychem Cable Accessories



Installation Instruction ESD-1426-GB-3/19

**Termination for B. R.
Polymeric Insulated Cables
for
300 mm² and 400 mm²
Concentric Cables**

25/44 kV

To view the TE Energy website:



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Before Starting

Check to ensure that the kit you are going to use fits the cable.

Refer to the kit label and the title of the installation instructions.

Components or working steps may have been modified since you last installed this product.

Carefully read and follow the steps in the installation instructions.

General Instructions

Use a propane (preferred) or butane gas torch.

Ensure the torch is always used in a well-ventilated environment.

Adjust the torch to obtain a soft blue flame with a yellow tip.

Pencil-like blue flames should be avoided.

Keep the torch aimed in the shrink direction to preheat the material.

Keep the flame moving continuously to avoid scorching the material.

Clean and degrease all parts that will come into contact with adhesive.

If a solvent is used follow the manufacturer's handling instructions.

Start shrinking the tubing at the position recommended in the instruction.

Ensure that the tubing is shrunk smoothly all around before continuing along the cable.

Tubing should be smooth and wrinkle free with inner components clearly defined.

The Information contained in these installation instructions is for use only by installers trained to make electrical power installations and is intended to describe the correct method of installation for this product. However, TE Connectivity has no control over the field conditions which influence product installation.

It is the user's responsibility to determine the suitability of the installation method in the user's field conditions.

TE Connectivity's only obligations are those in TE Connectivity's standard Conditions of Sale for this product and in no case will TE Connectivity be liable for any other incidental, indirect or consequential damages arising from the use or misuse of the products.

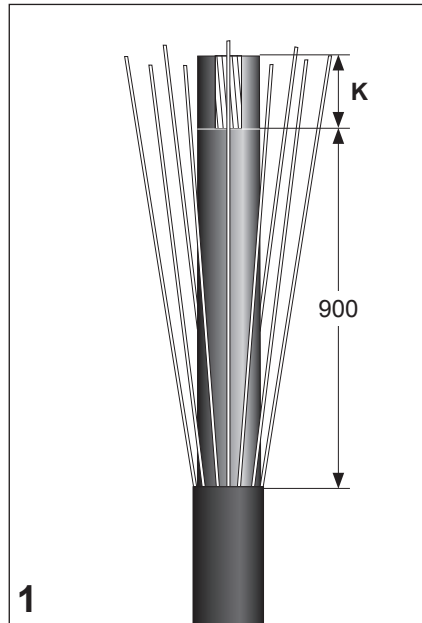
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Installation instruction for BR 300 mm² 1c 25/44 kV cables

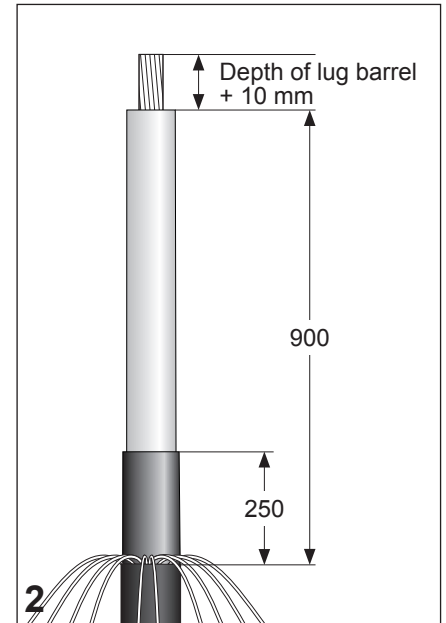
NOTE

This instruction is for use instead of installation instruction **EPP-0403** and is only to be used where the cable is 300 mm² or 400 mm² concentric neutral conductors.



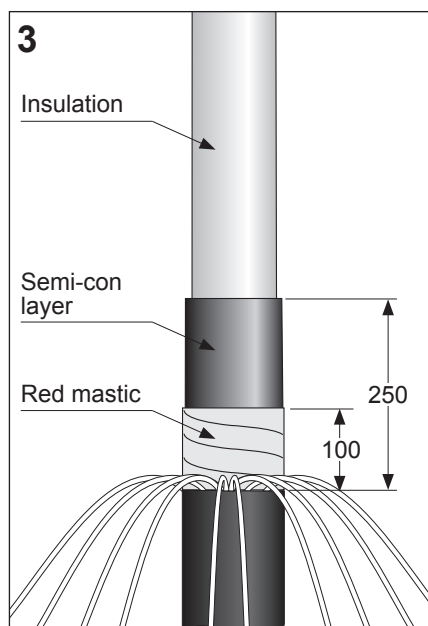
Strip the cable to the measurements given in the drawing.

Bend back the neutral wires.

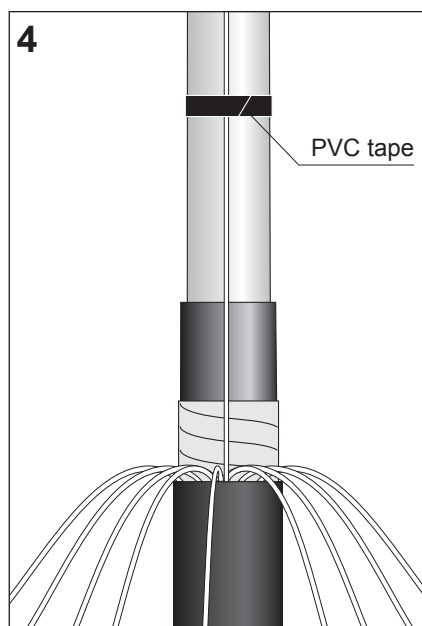


Remove the semi-conducting screen to 250 mm from the oversheath cut position with the approved stripping tool.

Wrap one layer of red mastic tape for 100 mm from the oversheath cut position using only a small overlap.

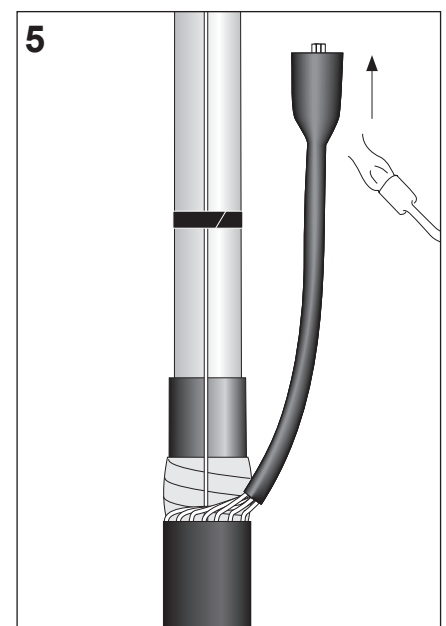


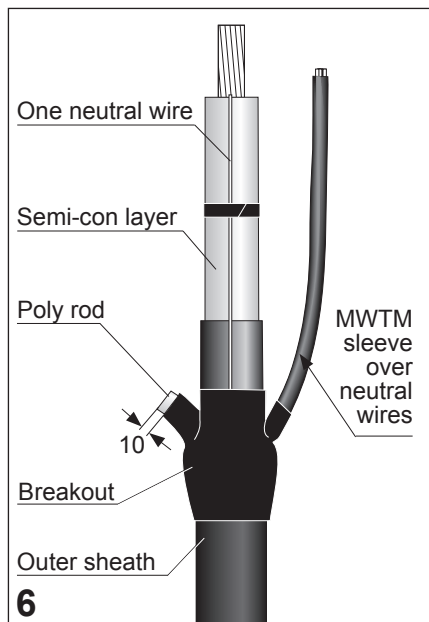
Take one neutral wire and tape it temporarily to the core insulation with PVC tape.



Form the remaining neutral wires into one conductor taking into account the neutral connection point.

Place the uncoated long black insulating sleeve from the additional kit over the neutral wires down into the crutch of the cable and shrink into place.



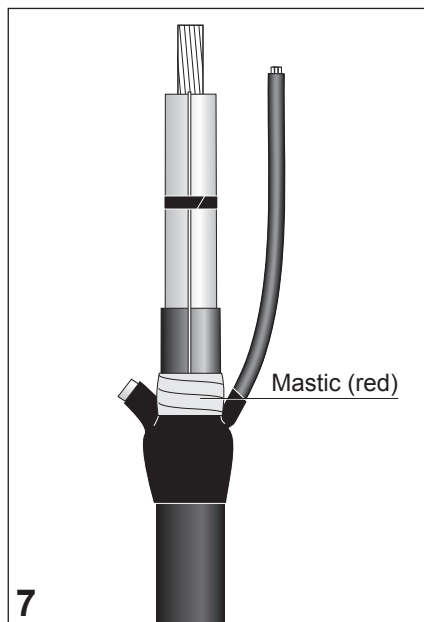


Position the cable breakout over the central core and the insulated neutral wires as far down into the crutch as possible.

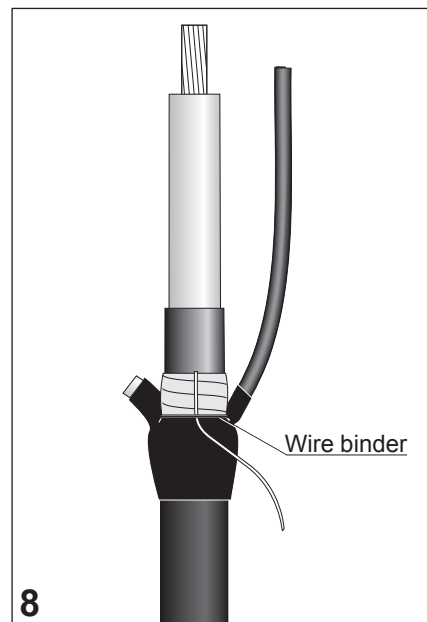
Place the poly rod in the third finger of the breakout allowing 10 mm to protrude.

Shrink the breakout into place ensuring a good seal onto the poly rod.

Allow to cool.

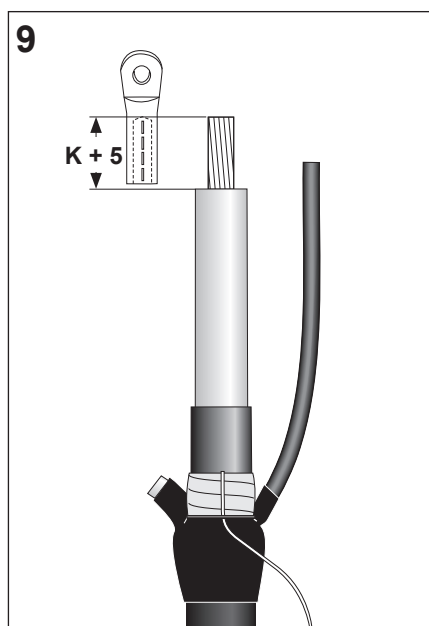


Wrap the central core finger of the breakout with red mastic tape for the full length of the breakout finger.

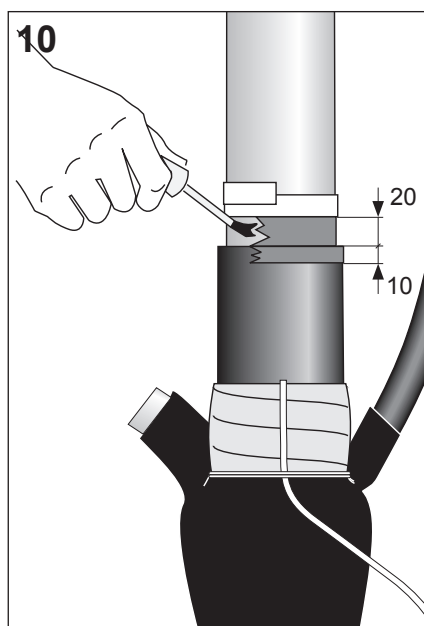


Bend the neutral wire down over the breakout finger and hold in place with a temporary wire binder.

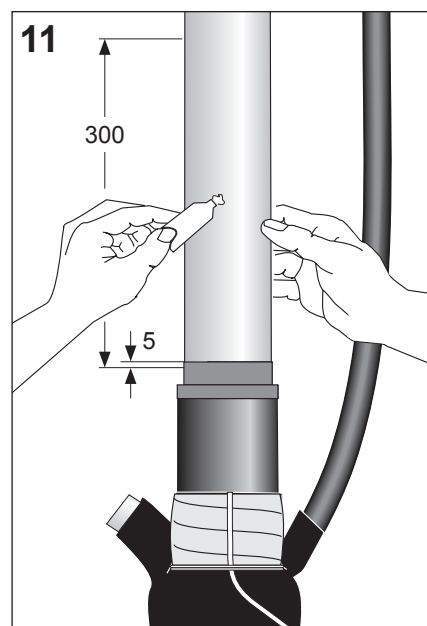
Cut back the insulation to $K = \text{depth of the lug barrel} + 5 \text{ mm}$.

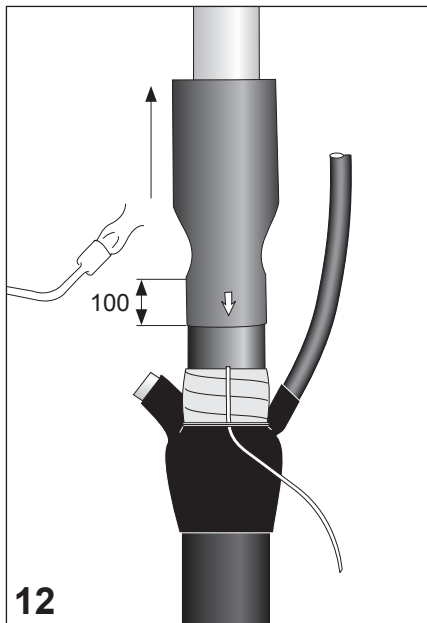


Apply the conductive paint; cover 10 mm of the semi-conductive layer and 20 mm of the insulation.



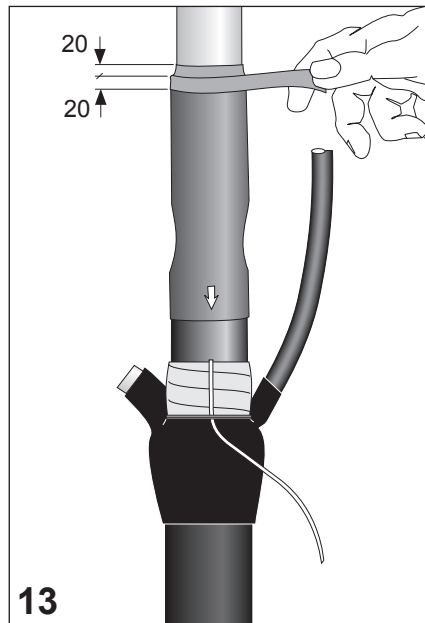
Apply the silicon grease; cover 5 mm of the conductive paint and 300 mm of the insulation.





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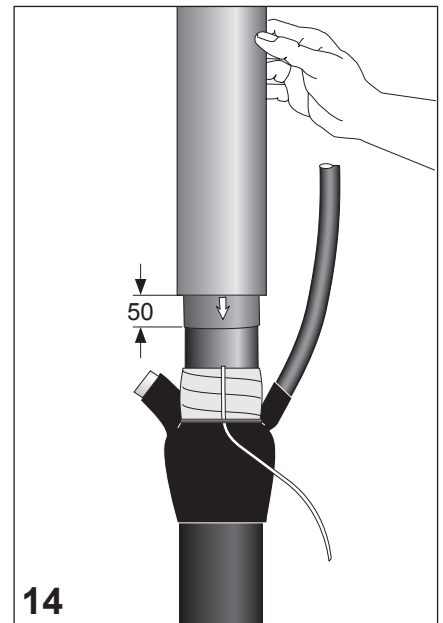
Position the long stress control tube with the arrow pointing downwards over the semi-conducting layer for 100 mm. Shrink down starting at the bottom and working upwards.



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Apply red mastic tape at the end of the stress control tube using slight tension and a small overlap.

Cover 20 mm of the stress control tube and 20 mm of the insulation.

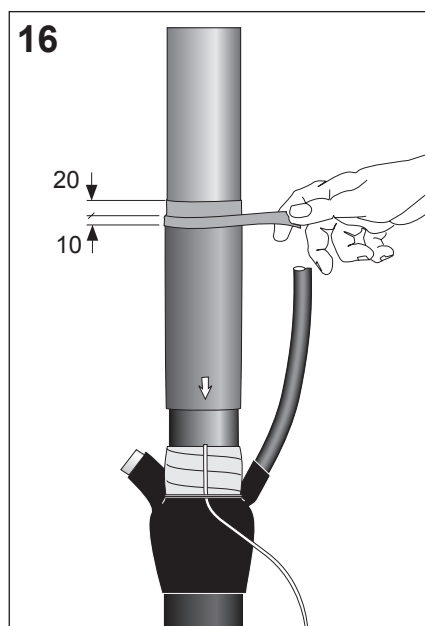
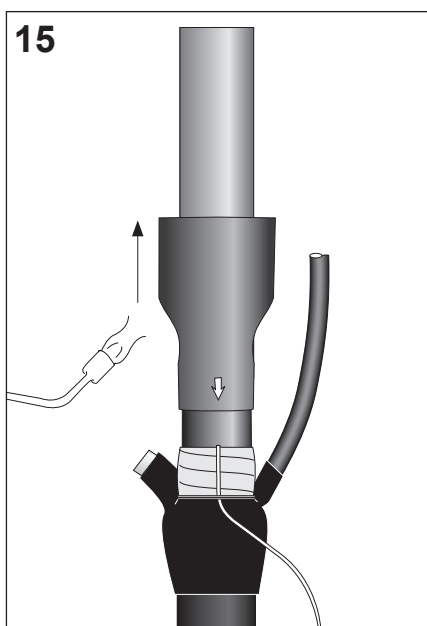


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Place the short red tube over the stress control tube leaving 50 mm of the stress control tube visible.

Shrink down starting at the bottom and working upwards.

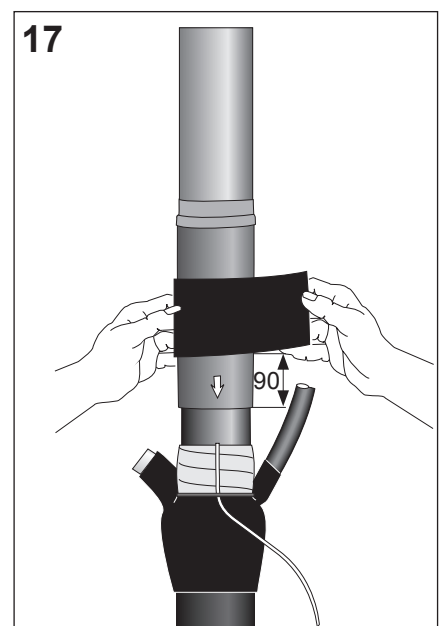
Place the short stress control tube over the core so that its end coincides with the end of the first stress control tube. Shrink down starting at the bottom and working upwards.



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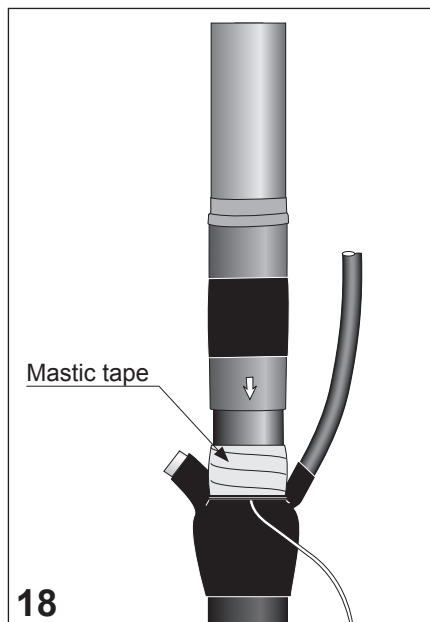
Apply red mastic tape at the top of the stress control tube.

Cover 10 mm of the stress control tube and 20 mm of the red tubing.



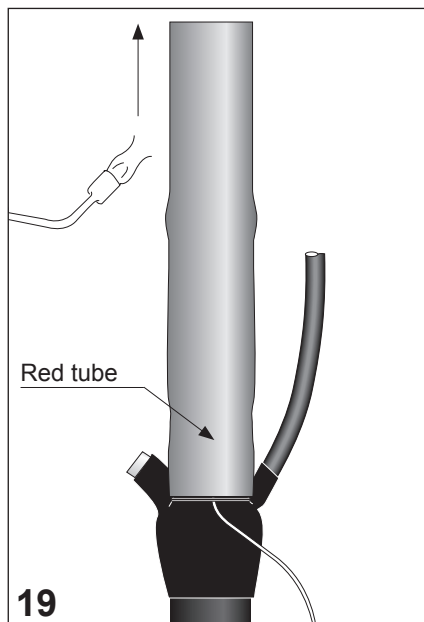
17

Wrap two layers of the black insulating profile over a length of approximately 120 mm starting 90 mm above the bottom end of the stress control tube.



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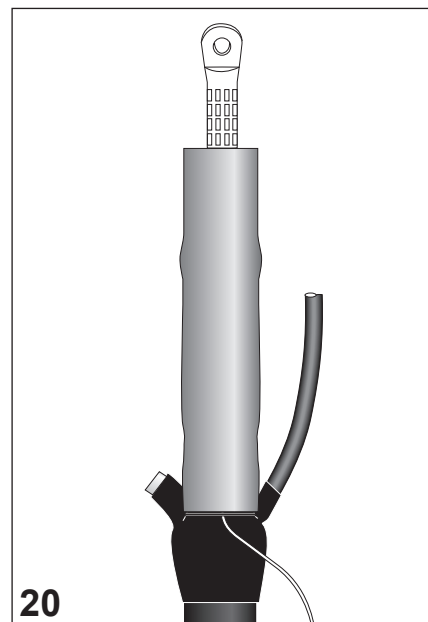
Wrap one layer of red mastic tape over the previously applied red mastic layer on the breakout finger of the phase conductor.



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Place the long red tube over the cable completely covering the red mastic layer on the breakout finger.

Shrink down starting at the bottom and working upwards.

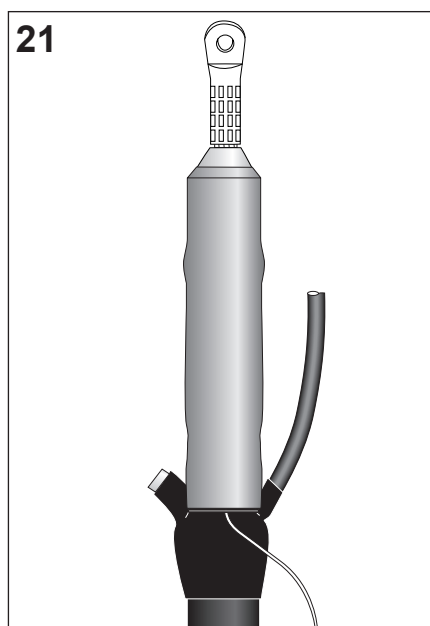


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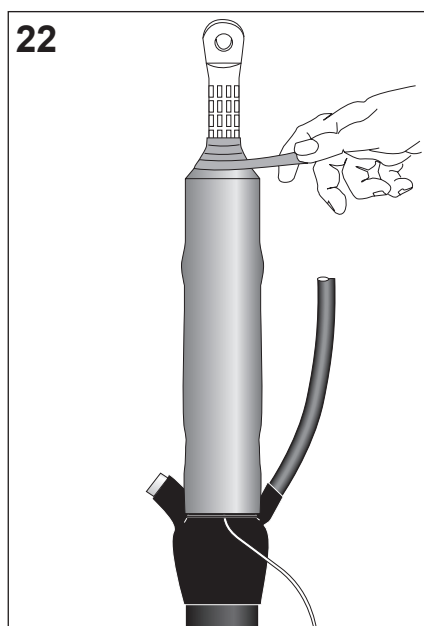
Cut the red tube back to the insulation cut.

Install, clean and degrease the cable lug.

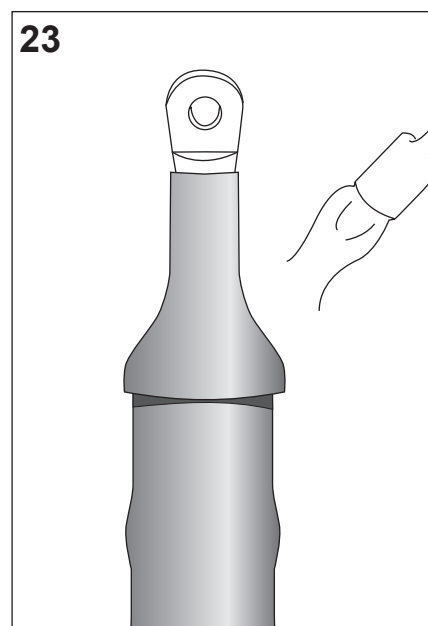
Chamfer the insulation to the diameter of the cable lug for a distance of approximately 30 mm.

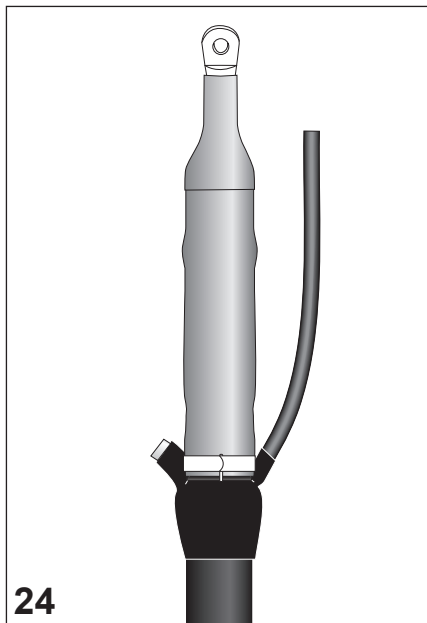


Wrap red mastic tape around the lug and the insulation, filling the gap between the lug and insulation to make a smooth profile.



Position the red sealing boot so that it covers the lug and the tube equally and shrink it into place starting at the top.

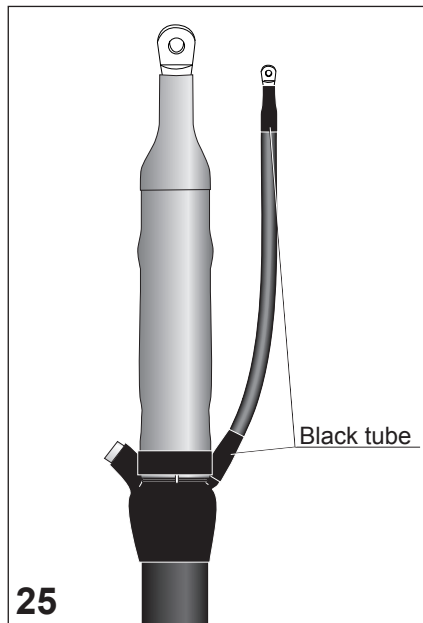




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Wrap the roll spring twice over the bottom of the red tube over the breakout finger. Bring the single neutral wire over the roll spring. Wrap the roll spring once over the neutral wire, then bring the neutral wire back over the roll spring. Wrap the rest of the roll spring over the neutral wire and tighten. Cut the excess length of wire close to the bottom of the roll spring.

Wrap 2 layers of PVC tape over the roll spring at the centre.



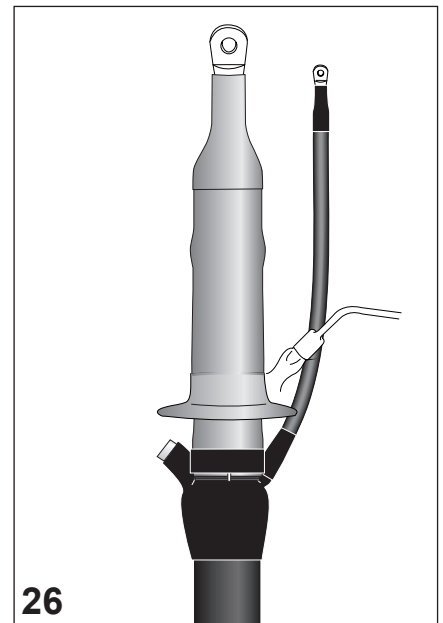
25

Place one short (150 mm) black tube over the insulated neutral wire conductor and over the breakout finger. Shrink into place to seal the breakout.

Cut the neutral wires to length and install the suitable lug. Position the other short black sealing sleeve over the lug and tube equally to seal the neutral conductors.

NOTE

The lug must be waterblocked.

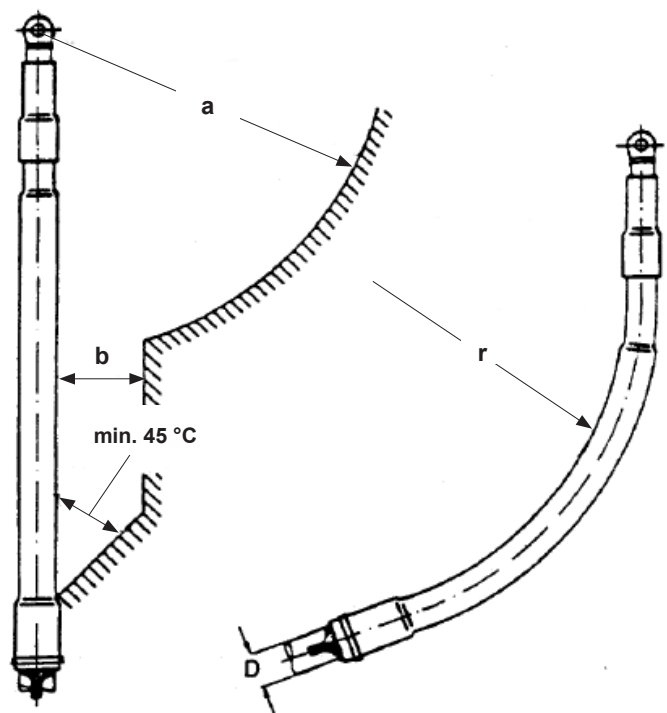
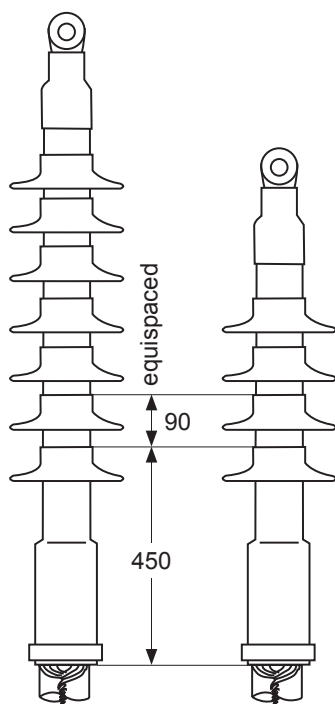


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Shrink the skirts into place according to the dimensions given in the drawing below.

Termination completed.

Please dispose of all waste according to environmental regulations.



Min. clearances	Max. system voltage 72 kV
a air clearance	according to local specifications (e. g. NBN, IEC, VDE)
b between ph/ph and ph/ground [mm]	100
r min. bending radius $10 \times D$, before bending heat cable up to approx. 70 °C	